COURSE 9A – Designing Web Pages Part 1(HTML and CSS)

9.1 Objectives

- To get introduced with Hyper Text Markup Language (HTML)
- To get introduced with Cascading Style Sheet (CSS).

9.2 HTML

9.2.1 What is HTML?

To understand HTML, let us first see the meaning of a Markup language

A markup language is a modern system for annotating a document in a way that is syntactically distinguishable from the text.

See example as indicated below:

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE recipe PUBLIC "-//Happy-Monkey//DTD RecipeBook//EN"
"http://www.happy-monkey.net/recipebook/recipebook.dtd">

<recipe>

    <title>Peanut-butter On A Spoon</title>

    <ingredientlist>
        <ingredient>Peanut-butter</ingredient>
    </ingredientlist>

    <preparation>
        Stick a spoon in a jar of peanut-butter,
        scoop and pull out a big glob of peanut-butter.
    </preparation>

</recipe>
```

The blue text shown in the example is the Markups to indicate the content enclosed in it.

HTML or HyperText Markup Language is the main markup language for creating web pages and other information that can be displayed in a web browser. It is a markup language that web browsers use to interpret and compose text, images and other material into visual or audible web pages.

Now it is simple. Isn't it? We may summarize the knowledge of HTML as below

- HTML stands for Hyper Text Markup Language
- HTML is a markup language
- A markup language is a set of markup tags
- The tags describe document content
- HTML documents contain HTML tags and plain text
- HTML documents are also called web pages

A HTML document when composed of the markup elements is saved as a file with a .html or .htm extension. The rendering of the document is done on any web browser such as Internet Explorer, Google Chrome, etc.,.

9.2.2 HTML Elements

HTML is written in the form of HTML elements consisting of tags enclosed in angle brackets (like <html>), within the web page content. HTML tags most commonly come in pairs like <h1> and </h1>, although some tags represent empty elements and so are unpaired, for example <img>. The first tag in a pair is the start tag, and the second tag is the end tag (they are also called opening tags and closing tags). In between these tags web designers can add text, further tags, comments and other types of text-based content. Most HTML elements have attributes.

The browser does not display the HTML tags, but uses the tags to interpret the content of the page. The purpose of a web browser is to read HTML documents and compose them into visible web pages.

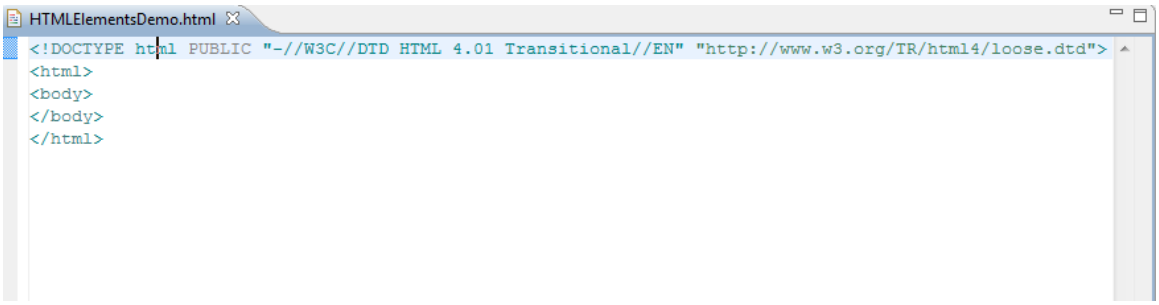
Let us see each of the HTML element in detail:

DOCTYPE

The <!DOCTYPE> declaration helps the browser to display a web page correctly.

There are many different documents on the web, and a browser can only display an HTML page 100% correctly if it knows the HTML type and version used

Below is a HTML document showing the <!DOCTYPE> declaration in the first line.



HTML

The <html> element defines the whole HTML document. The element has a start tag <html> and an end tag </html>.

BODY

The <body> element defines the body of the HTML document. The element has a start tag <body> and an end tag </body>.

Headings

HTML headings are defined with the <h1> to <h6> tags

HTMLElementsDemo.html

```
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">
<html>
<body>
<h1>My Heading</h1>
<h2>My Heading</h2>
<h3>My Heading</h3>
<h4>My Heading</h4>
<h5>My Heading</h5>
<h6>My Heading</h6>
</body>
</html>
```

Output of the above code may be viewed using a web browser as below:

My Heading

My Heading

My Heading

My Heading

My Heading

My Heading

Various other elements in HTML

The other basic elements that are widely used in HTML are tabulated as below

Element	Description
<p>	Defines a paragraph in the HTML document
 	Defines a line break
<hr>	Defines a horizontal line
<!-->	Defines a comment in the HTML document and is not interpreted by browser

Sample code and its output:

HTMLElementsDemo.html

```
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">
<html>
<body>
<h1>My Heading</h1>
<p>My Paragraph</p>
<br></br>
<p>My paragraph content to be placed here</p>
<hr></hr>
<h1>Your Heading</h1>
<p>Your Paragraph</p>
<br></br>
<p>Your paragraph content to be placed here</p>
<hr></hr>
</body>
</html>
```

My Heading

My Paragraph

My paragraph content to be placed here

Your Heading

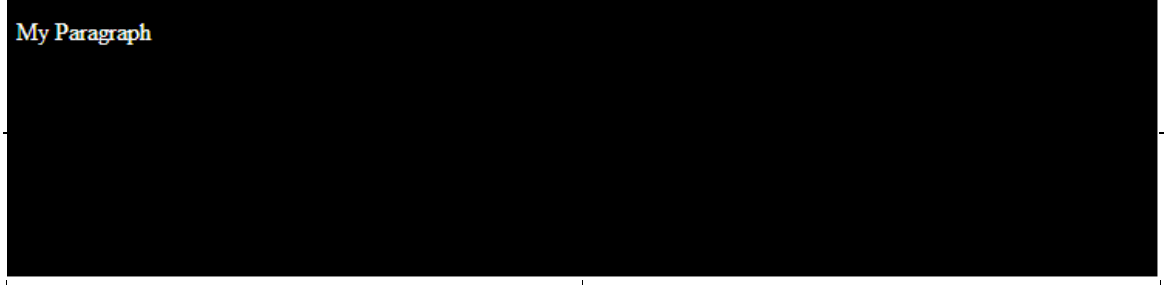
Your Paragraph

Your paragraph content to be placed here

9.2.3 HTML Head elements

The elements of the head element with the associated nested tags are listed in the table below

Element	**Description**
<head>	Defines information about the document
<title>	Defines the title of a document
<style>	Defines style information for a document
<link>	Defines the relationship between a document and an external resource
<meta>	Defines metadata about an HTML



	document
<script>	Defines a client-side script

HTMLElementsDemo.html HTMLHeadDemo.html

```
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">
<html>
<head>
<meta name="keywords" content="HTML, headTag, title, link, script">
<title>HTML Head Demo</title>
<link rel="stylesheet" type="text/css" href="mystyle.css">

<script type="text/javascript">
</script>

<style type="text/css">
body {background-color:black;}
p {color:white;}
</style>

</head>
<body>
<p>My Paragraph</p>
</body>
</html>
```

The sample HTML document consisting of these head elements is shown here with associated output screen

9.2.4 HTML formatting elements

The HTML formatting elements have been tabulated here.

Element	**Description**
<b>	Defines bold text
<i>	Defines in italics
<small>	Defines smaller text

	Defines important text
<sub>	Defines subscripted text
<sup>	Defines superscripted text

Sample code for some of the formatting tags are given below. The output has also been displayed along with.

HTMLElementsDemo.html

```
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">
<html>
<body>
<h1>My Heading</h1>
<p><b>My Paragraph</b></p>
<br></br>
<p><small>My paragraph <i>content</i> to be placed here</small></p>
<hr></hr>
<h1>Your Heading</h1>
<p><b>Your Paragraph</b></p>
<br></br>
<p><strong>Your paragraph <i>content</i> to be placed here</strong></p>
<hr></hr>
</body>
</html>
```

My Heading

My Paragraph

My paragraph *content* to be placed here

Your Heading

Your Paragraph

Your paragraph *content* to be placed here

9.2.5 HTML Attributes

Attributes in HTML element provide additional information about the element.

Attributes	**Description**
class	Specifies one or more classnames for an element (refers to a class in a style sheet)
id	Specifies a unique id for an element
style	Specifies an inline CSS style for an element
title	Specifies extra information about an element (displayed as a tool tip)

9.2.6 HTML Links

The HTML <a> tag defines a hyperlink.

A hyperlink (or link) is a word, group of words, or image that you can click on to navigate to another document.

When you move the cursor over a link in a Web page, the arrow will turn into a little hand. The most important attribute of the <a> element is the href attribute, which indicates the link's destination.

By default, links will appear as follows in all browsers:

- An unvisited link is underlined and blue
- A visited link is underlined and purple
- An active link is underlined and red

Samples for defining HTML link are shown here:

HTMLElementsDemo.html | HTMLHeadDemo.html

```
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">
<html>
<head>
<meta name="keywords" content="HTML, headTag, title, link, script">
<title>HTML Head Demo</title>
<link rel="stylesheet" type="text/css" href="mystyle.css">
<script type="text/javascript">
</script>
<style type="text/css">
body {background-color:white;}
p {color:black;}
</style>
</head>
<body>
<h4><a href="http://www.google.com/">Search</a>
</h4>
<p><a id="paraId">My Paragraph<br/>
Lorem ipsum dolor sit amet, consectetur adipiscing elit
</a></p>
</body>
</html>
```

Search

My Paragraph
Lorem ipsum dolor sit amet, consectetur adipiscing elit

9.2.7 HTML Images

In HTML, images are defined with the <img> tag. The <img> tag is empty, which means that it contains attributes only, and has no closing tag. Some of the attributes of the image tag are

- src – Indicates the source or the URL of the image that is to be displayed
- alt – alternate name or text for the image
- height, width – attributes describing the size

Below is an example (image will be visible if it is placed in same directory where we have html page created and image file is named as smiley.gif):

HTMLElementsDemo.html | HTMLHeadDemo.html

```
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">
<html>
<head>
<meta name="keywords" content="HTML, headTag, title, link, script">
<title>HTML Head Demo</title>
<link rel="stylesheet" type="text/css" href="mystyle.css">
<script type="text/javascript">
</script>
<style type="text/css">
body {background-color:white;}
p {color:black;}
</style>
</head>
<body>

<img src="smiley.gif" alt="Smiley face" width="42" height="42">

</body>
</html>
```

9.2.8 HTML Division

The tag defines a division or a section in an HTML document. The tag is used to group block-elements to format them with CSS. The attributes are

class and style.

Note: The element is very often used together with CSS, to layout a web page.

A Simple Example to show the usage:

HTMLElementsDemo.html | HTMLHeadDemo.html

```
<!DOCTYPE html PUBLIC "-//W3C//DTD HTML 4.01 Transitional//EN" "http://www.w3.org/TR/html4/loose.dtd">
<html>
<head>
<meta name="keywords" content="HTML, headTag, title, link, script">
<title>HTML Head Demo</title>
<link rel="stylesheet" type="text/css" href="mystyle.css">
<script type="text/javascript">
</script>
<style type="text/css">
body {background-color:white;}
p {color:black;}
</style>
</head>
<body>
<div>
<img src="smiley.gif" alt="Smiley face" width="42" height="42">
</div>
<div>
<p>Hello!! Have a Great Day</p>
</div>
</body>
</html>
```

Hello!! Have a Great Day

9.2.9 HTML LIST

HTML provides the means for producing three types of lists: unordered (ie., unnumbered) and ordered (ie., numbered) lists, Definition (I.e define the terms like in glossary).The UL, OL and DL elements are block elements.

Unordered Lists:

An unordered list typically is a bulleted list of items. This is probably the most common type of list on the Web. The <UL> tag opens an unordered list while </UL> closes it. Between these tags are placed list items with an <LI> tag as follows:

type : Determines the form of bullet which precedes the list item placed within an unordered list. Values are disc,square,circle

*ListDemo.html

```
<UL type="circle">
<LI>red</LI>
<LI>yellow</LI>
<LI>blue</LI>
</UL>
```

Output

- red
- yellow
- blue

Ordered Lists:

An ordered list is formatted exactly the same as an unordered list, except that <OL> tags are used instead of <UL>. In an ordered list, sequential numbers are generated automatically, as shown below:

type attribute Determines the form of bullet which precedes the list item placed within an unordered list. Values are 1,a, A,i,I.

*ListDemo.html

```
<OL type="1">
<LI>red</LI>
<LI>yellow</LI>
<LI>blue</LI>
</OL>
```

Output

1. red
2. yellow
3. blue

Definition

This contains a series of terms and definitions, and would typically be used in a glossary.

*ListDemo.html

```
<dl>
  <dt>Sump</dt>
  <dd>A place where water completely fills
  the cave passage</dd>
  <dt>Rift</dt>
  <dd>A vertical fracture in the rock, created
   by geological stress</dd>
  <dd>A passage formed along such a fracture,
  usually tall and narrow</dd>
  <dt>Abseil</dt>
  <dt>Rappel</dt>
  <dd>To descend a rope using a device to control
  speed</dd>
</dl>
```

Output:

Sump

A place where water completely fills the cave passage

Rift

A vertical fracture in the rock, created by geological stress

A passage formed along such a fracture, usually tall and narrow

Abseil

Rappel

To descend a rope using a device to control speed

9.2.10 HTML TABLE

A table is created using the <table> tag and a closing </table> tag. A table consists of rows and columns.

Attributes of the <table> tag

•border - takes a numeric value denoting the thickness of the border around the table.

•cellspacing - takes a numeric value denoting how much space to put between cells.

•cellpadding - takes a numeric value denoting how much padding to put between what is in the cells and the cell walls.

For example

```
*TableDemo.html
<html>
<head>
<title>HTML Table Demo</title>
</head>
<body>
<table border="1" cellspacing="2" cellpadding="2">
<!-- table content goes here -->
</table>
</body>
</html>
```

Table rows

Table rows are created using the <tr> tag which should go inside the <table> tag. Use an opening <tr> tag and its closing tag </tr> for every row in the table.

for example

```
*TableDemo.html
<table border="1" cellspacing="2" cellpadding="2">
<tr></tr>
<tr></tr>
</table>
```

Table cells

Table cells are created using the <td> tag which should go inside the <tr> tag. Use an opening <td> tag and its closing tag </td> for every cell in the table.

```
*TableDemo.html
<table border="1" cellspacing="2" cellpadding="2">
<tr>
<td></td>
<td></td>
</tr>
<tr>
<td></td>
<td></td>
</tr>
</table>
```

Adding content to the table

Now that we have a structure for our table, we can add some content to it. The content of a table is placed inside the table's cells..so inside the <td> tags in the table.

```
*TableDemo.html
<table border="1" cellspacing="2" cellpadding="2">
<tr>
<td>First row, first column</td>
<td>First row, second column</td>
</tr>
<tr>
<td>HTML</td>
<td>Javascript</td>
</tr>
<tr>
<td>CSS</td>
<td>PHP</td>
</tr>
</table>
```

Output

First row, first column	First row, second column
HTML	Javascript
CSS	PHP

Also you can add headings & captions to tables, male cells span multiple rows & columns, align the content within cells, and more.

Empty cells

Sometimes you may not need to place content inside of a table cell, instead you just need the cell to be there. But a table cell may not appear unless it has content inside of it.

You can include empty cells in a table by placing the   character entity inside a table cell.

```
*TableDemo.html
<table border="1" cellspacing="2" cellpadding="2">
<tr>
<td>Four cells in this table</td>
<td>One is empty</td>
</tr>
<tr>
<td>Three are filled</td>
<td> </td>
</tr>
</table>
```

Output

Four cells in this table	One is empty
Three are filled	

Headings

You can display headings in a table using the <th> tag. A heading is an emphasized cell in a table. Use the <th> tag just like the <td> tag.

```
*TableDemo.html
<table border="1" cellspacing="2" cellpadding="2">
<tr>
<th>Web development</th>
<th>Software development</th>
</tr>
<tr>
<td>HTML</td>
<td>C++</td>
</tr>
<tr>
<td>PHP</td>
<td>Fortran</td>
</tr>
</table>
```

Output

Web development	Software development
HTML	C++
PHP	Fortran

Captions

A caption is added to a table using the <caption> tag which should be placed right below the <table> tag before the first <tr> tag.

TableDemo.html

```html
<table border="1" cellspacing="2" cellpadding="2">
<caption>Computer languages</caption>
<tr>
<th>Web development</th>
<th>Software development</th>
</tr>
<tr>
<td>HTML</td>
<td>C++</td>
</tr>
<tr>
<td>PHP</td>
<td>Fortran</td>
</tr>
</table>
```

Output

Computer languages

Web development	Software development
HTML	C++
PHP	Fortran

Aligning table content

You can align table content using the align attribute inside the <td> tag. This attribute can take the values

left - to align content to the left

center - to align content to the center

right - to align content to the right.

Just as you can set the alignment of table content, you can set the alignment of the table itself. This is done the same way as setting the alignment of table content.

Spanning multiple/rows columns

Table column spanning

Why should all rows have the same number of columns in a table? There may be situations where you need a cell or cells to span more than one column. This is achieved using the colspan attribute of the <td> tag. The colspan attribute takes a numeric value indicating how many columns a cell should span.

*TableDemo.html

```
<table border="2" cellspacing="4" cellpadding="10"
width="30%" align="left">
<caption>Web Languages</caption>
<tr>
<th width="40%" align="center">Client-side</th>
<td align="center">HTML</td>
<td align="center">CSS</td>
<td align="center">Javascript</td>
</tr>
<tr>
<th align="center">Server-side</th>
<td align="center">ColdFusion</td>
<td align="center" colspan="2">PHP</td>
</tr>
</table>
```

Output

Web Languages

Client-side	HTML	CSS	Javascript
Server-side	ColdFusion	PHP	

Table row spanning

You can make a cell span more than one row using the rowspan attribute of the <td> tag. The rowspan attribute takes a numeric value indicating how many rows it should span.

```
*TableDemo.html
<table border="2" cellspacing="4" cellpadding="10"
width="30%" align="left">
<caption>Web Languages</caption>
<tr>
<th align="center" rowspan="2">Web languages</th>
<td align="center">HTML</td>
<td align="center">CSS</td>
</tr>
<tr>
<td align="center">Javascript</td>
<td align="center">PHP</td>
</tr>
<tr>
<tr>
<th align="center" rowspan="2">Software languages</th>
<td align="center">C</td>
<td align="center">C++</td>
</tr>
<tr>
<td align="center">FORTRAN</td>
<td align="center">Visual Basic</td>
</tr>
<tr>
</table>
```

Output

Web Languages

Web languages	HTML	CSS
	Javascript	PHP
Software languages	C	C++
	FORTRAN	Visual Basic

9.2.11 HTML FORMS

HTML Forms are required when you want to collect some data from the user and process it.

For example registration information: Name, address, date of birth, contact number, mail-id etc.

The <form> tag is used to create an HTML form:

```
<body>
<form>
..
input elements
..
</form>
</body>
```

A form will take input from the user using different form controls and form elements and then the data is passed to server for processing.

- Users interact with forms through “form controls”.
- Each control has both an initial value and a current value.
- A control's *initial value* may be specified with the control element's ”value” attribute.
- The control's *current value* is first set to the initial value. Thereafter, the control's current value may be modified through user interaction and scripts.
- A control's initial value does not change. Thus, when a form is reset, each control's current value is reset to its initial value.

HTML defines the following control types:

Buttons

HTML provides three types of buttons:

- Submit buttons: When activated, a submit button submits a form.A form may contain more than one submit button.
- Example for submit button:

```
<form name="input" action="demo_form_action.asp" method="get">
Username: <input type="text" name="user">
<input type="submit" value="Submit">
</form>
```

- <input type="submit" value="submit"> is used to create submit buttons.
- reset buttons: When activated, a reset button resets all controls to their initial values.
- <input type="reset" value="reset"> is used to create submit buttons.
- push buttons: Push buttons have no default behavior. Each push button may have scripts associated with the element's events attributes. When an event occurs (e.g., the user presses the button, releases it, etc.), the associated script is executed. e.g:<button name="edit" value="edit"> is used to create buttons.

Checkboxes

- Checkboxes are on/off switches that may be toggled by the user.
- The defaultvalue of checkbox control is off, but when user selects a checkbox, the control becomes on.
- "checked" attribute can be used when we want a default value to be selected.
- Checkboxes allow users to select several values from the set. The *input* element is used to create a checkbox control.
- Example:

```
<form name="input" action="demo_form_action.asp" method="get">
<input type="vehicle" value="Bike">I have a Bike<br>
<input type="vehicle" value="Car">I have a Bike
</form>
```

How the HTML code above looks in a browser:

☐I have a bike
☐I have a car

Radio buttons :

- Radio buttons are also on/off switches that may be toggled by the user.

- They are similar to checkboxes, but only one option can be selected from the set.
- e.g : <input type=”radio” name=”color” value=”red”><input type=”radio” name=”color” value=”blue”> can be used to create a radio button control with two values as red and blue.
- Example:

```
<form>
<input type="radio" name= "sex" value="male">I have a Bike<br>
<input type="radio" name ="sex" value="Female">I have a Bike
</form>
```

How the HTML code above looks in a browser:

Male
Female

Menus

- Menus are used when we want to choose a option from set of available options.
- The *select* element is used to create a menu, and option element is used to define options of that menu.
- e.g: <select> <option> red </option> <option> blue </option> </select> can be used to create a menu with two options as red and blue.

Text Fields & Textareas

When it comes to text controls within forms there are a few different elements available to obtain data. Text fields and textareas are specifically used to gather text or string based data. This may include data containing passages of text content, passwords, telephone numbers, and so forth.

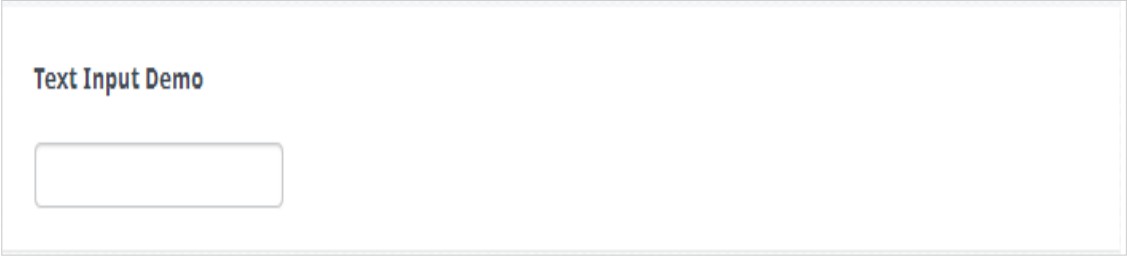
Text Fields

One of the primary elements used to obtain text from users is the input element. The input element uses the type attribute to determine what type of information is to be captured within the specific control. The most popular type attribute value is text, which denotes a single line text input.

Along with setting a type attribute it is also best practice to give an input a name attribute as well. The name attribute is used as the name of the control and is submitted

```
<input type ="text" name="sample_text_field">
```

along with the input's data to the server.



Textarea

Another element used to capture text based data is the textarea element. The textarea element differs from the text input in that it is for larger passages of text spanning multiple columns. The textarea also has start and end tags which can wrap plain text. Since the textarea element only accepts one type of value the type attribute doesn't apply here, however the name attribute is still in effect.

```
<textarea name="sample_textarea">Sample textarea</textarea>
```

Text input

- HTML provides two types of controls that allow users to input text.
- The *input* element creates a single-line input control and
- *textarea* element creates a multi-line input control.
- Example:

```
<form>
First name: <input type="text" name="firstname"><br>
Last name: <input type="text" name="lastname">
</form>
```

The above HTML code looks in a browser as below:

First name:

Last name:

File select

This control type allows the user to select files so that their contents may be submitted with a form.

- The *input* element is used to create a file select control.

Drop Down Lists:

Drop down lists are a perfect way to provide users with a long list of options in a usable manner. Outputting every state within the country on a page with radio buttons would create a rather cluttered and daunting list. Drop down menus provide the perfect venue for a long list of options.

To create a drop down menu the select and option elements are used. The select element will wrap all of the different menu options marked up using the option element. Then you can apply the name attribute to the select element.

Each individual option within the menu needs to be coded using the option element. The option element will wrap the text to be included within the menu. Additionally, the option element will include the value attribute, specific to each individual option.

As with the checked Boolean attribute with radio buttons and checkboxes, drop down menus can use the selected Boolean attribute to preselect an option for users.

```
<select name="day">
  <option value="Friday" selected>Friday</option>
  <option value="Saturday">Saturday</option>
  <option value="Sunday">Sunday</option>
</select>
```

Drop Down Lists Demo

Friday

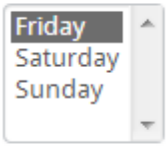
Multiple Selections

Using the standard drop down list and adding the Boolean attribute multiple to the select element allows a user to choose more than one option from the menu. Additionally, using the selected Boolean attribute on more than one option element within the menu will preselect multiple options.

The height and width of the select element can be controlled using CSS and should be adjusted appropriately to allow for multiple selections. It may also be worth mentioning to users that to choose multiple options they will need to hold down the shift key while clicking to make their selections.

```
<select name="day" multiple>
  <option value="Friday" selected>Friday</option>
  <option value="Saturday">Saturday</option>
  <option value="Sunday">Sunday</option>
</select>
```

Multiple Selections Demo



Form Buttons

After a user inputs the requested information, buttons allow them to put that information into action. Most commonly, a submit button is used to process the data. A reset button is often used to clear data.

Submit Button

Users click the submit button to process data after filling out a form. The submit button uses the input element with a type attribute of either submit or image. The submit attribute value is the most common as it is simple to use and provides the most control. The image attribute value is used specifically to set an image as a submit button, however with the use of CSS the need to use an image has greatly diminished.

To determine the verbiage to be used within the button, the value attribute is used. Using CSS properties such as background, border-radius, box-shadow, and others, the button can then be styled to any specific desire.

```
<input type="submit" name="submit" value="Submit Form">
```

Submit Button Demo

SUBMIT FORM

Reset Button

To take the complete opposite action from submitting a form, users may also reset a form using the reset button. The reset button code works just like that of the submit button, however it uses the reset value for the type attribute.

Reset buttons are becoming less and less popular because they offer a very strong action, often undesired by both websites and users. Users may spend quite a bit of time filling out a form only to click the reset button accidentally thinking it was the submit button. Chances of a user filling out the form again thereafter are small. Before using a reset button think of any potential consequences.

```
<input type="reset" name="reset" value="Reset Form">
```

Reset Button Demo

RESET FORM

Other Inputs

Outside all other previously stated choices the input element does have a few other use cases. Two of which include passing hidden data and attaching filings during form processing.

Hidden Input

Hidden inputs provide a way to pass data to the server without displaying it to users. Hidden inputs are typically used for tracking codes, keys, or other information not pertinent to the users but helpful to the website overall. This information is not displayed on the page, however it could be seen by viewing the source code of a page. That said, it should not be used for sensitive or secure information.

To create a hidden input you use the hidden value for the type attribute. Additionally, you pass along the appropriate name and value.

```
<input type="hidden" name="tracking_code" value="abc_123">
```

File Input

To allow users to add a file to a form, much like that of attaching a file to an email, the file value for the type attribute is used. The file input is most commonly seen to upload pictures or files to social networks or applications.

Unfortunately, stylizing the file input is a tough task with CSS. Each browser has its own default rendering of how the input should look and doesn't provide much control to override the default styling. JavaScript and other solutions can be built to allow for file input, but they are slightly more difficult to construct.

```
<input type="file" name="file">
```

File Input Demo

Choose File No file chosen

Organizing Form Elements

Knowing how to capture data with inputs is half the battle. Organizing form elements and controls into a usable manner is the other half. Forms are not worth much unless users understand what is asked of them and how to provide the requested information. Using labels, fieldsets, and legends we can better organize forms and guide users to completing the end task.

Label

Labels provide captions, or headings, for form elements. Created using the label element, labels should include descriptive text, of which describes the input or control it pertains to. Labels should include a for attribute. The value of the for attribute should be the same as the value of the id attribute included within the form element the label corresponds to. Matching up the for and id values ties the two elements together, allowing users to click on the label to get focused within the proper form element.

```
<label for="username">Username</label>
<input type="text" name="username" id="username">
```

Label Demo

Username

When using labels with radio buttons or checkboxes the input element can be wrapped within the label element. Doing so allows omission of the for and id attributes.

```
<label><input type="radio" name="day" value="Friday" checked> Friday</label>
<label><input type="radio" name="day" value="Saturday"> Saturday</label>
<label><input type="radio" name="day" value="Sunday"> Sunday</label>
```

Label Radio Button & Checkbox Demo

Friday Saturday Sunday

Fieldset

Fieldsets group form controls and labels into organized sections. Much like a div or other structural element, the fieldset is a block level element that wraps related elements, however specifically within a form for better organization. Fieldsets by default also include a border outline that can be modified using CSS.

```
<fieldset>
  <label for="username">Username</label>
  <input type="text" name="username" id="username">
  <label for="password">Password</label>
  <input type="text" name="password" id="password">
</fieldset>
```

Fieldset Demo

Username Password

9.3 CSS

9.3.1 Introduction

Css stands for cascading style sheets. Styles define how html tags are displayed or formatted in the browser. Styles can be applied to html tags to format the look and feel of the html pages. By defining CSS based styling, the styling/formatting code can be separated from html code.

Styles are normally defined in an external file called style sheets which have a .css extension. The stylesheets are then imported into the html file. CSS works by defining rules how a selector (html tag) can be displayed.

9.3.2 CSS Syntax

A CSS rule set consists of a selector and a declaration block. The general syntax of a CSS rule set looks like this:

Selector {property:value; property:value; . . .}

- Selector – points to the html element that needs to be formatted
- Property – points to the specific html element's attribute that is being formatted
- Value – is the value given to the attribute that is used for formatting

More than one attribute/property of the html element can be specified in the CSS ruleset. The different sets of properties/value pairs are separated by a semi-colon. To improve the readability the property:value pairs can be written on separate lines.

p{color:blue; text-align:left;}

The ruleset above formats the color of the text to blue inside a html <p> tag and aligns the text to left.

p

{

color:red;

font-style:italic;

font-family:"arial";

}

The ruleset above formatted the color of text to red inside a html <p> tag, and formats the text to italics and sets to Arial font.

9.3.3 Grouping Selectors

In style sheets there are often elements with same styling. To minimize code the selectors can be grouped. To group selectors separate each selector with a comma.

Objective: Group selectors to apply same styling to multiple html elements

```
h2, h3, p
{
 color:grey;
 font-weight:italics;
 font-family:verdana;
}
```

html

```html
<html>
    <head>
          <Title>Content Alignment and CSS</Title>
          <link href='style.css' type='text/css' rel='stylesheet' />
    </head>
    <body>
          <div class='container'>
          <div class='left'>
                <h2>Few Indian Beverages</h2>
                <ul>
                      <li id='first'>Lassi</li>
                      <li>Chhaas</li>
                      <li>Aamras</li>
                      <li>Sugarcase juice</li>
                </ul>

          </div>
          <div class='right'>
                <h3>Popular Indian Beverages</h3>
                <ul>
                      <li style='color: green'>Coffee</li>
                      <li>Tea</li>
                      <li>Jal-jeera</li>
                      <li>Sharbat</li>
                </ul>
          </div>
          </div>
          <p>Above examples are taken for demonstrating the usage of style sheets and
          its implementation.</p>
    </body>
</html>
```

9.3.4 More About CSS Selectors

The selectors can be used to find all or specific html elements and apply styling to a single html element or to more than one or the entire html elements at once.

Id selector

The Id selector uses the “id” attribute of an html tag to find the specific html element. The id should be a unique in an html page. Using the unique id then you can find a single html element and apply styling.

In order to use the id selector, first the html element is given a unique id. The id can be a character, string or a combination of characters and numbers. The id should not start with a number. Then to find the element using the id write hash character followed by the id of the element.

```
#p1,h1
{
 text-align: left;
 color:red;
}
```

The ruleset above will find the html para and heading elements with the specified id values and apply the styling.

Class selector

The class selector uses the html “class” attribute of an html tag to find the html elements. Unlike “id” selector “class” is generally not given a unique value. Thus the “class” attribute can be used to apply styling to more than one html element that have the same value for the class attribute.

In order to use the class selector, first the html element is given a class name. The name of the class can be a character, string or a combination of characters and numbers. The class name cannot start with a number. Then to find the emelemt using the class name write a period character followed by the class name.

```
p.center, tr.center
{
```

```
 text-align: center;
 color:red;
}
```

The ruleset above finds all the html para and table row elements with class=”center” and applies the styling.

A class selector can be created free of a tag name in which case all the tags that have the same class name will be styled similarly.

```
.right
{
text-align:right;
color:blue;
font-family:”calibri”;
}
```

The id selector can be applied to one , unique element compared to the class selector which can be applied to several different elements.

Objective: To apply style using id *and class attributes.*

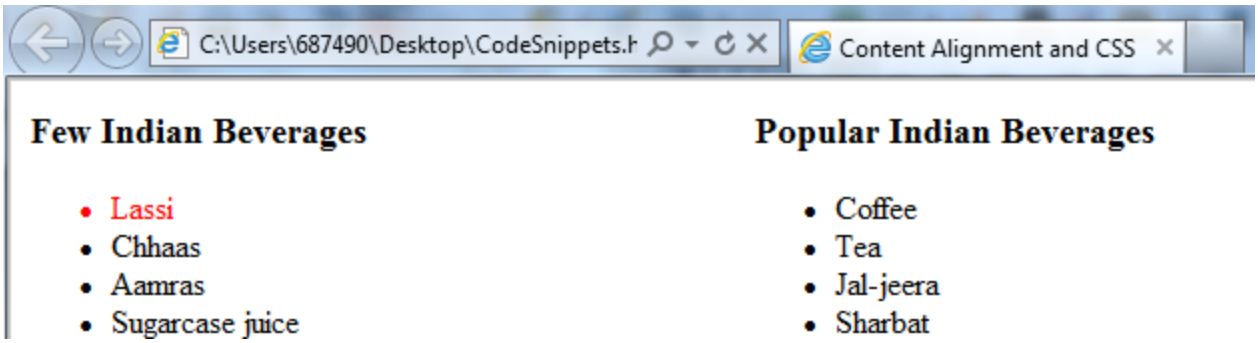
'.' in front of each tag defines a class and corresponding class is mentioned in the element used in HTML page.

We can use '#' to define style for a id and give the corresponding id for the element in HTML page. Id for each element in HTML page is always unique. No two elements in an HTML file should have the same id.

```
.left {
  width: 39%;
  float:left;
  text-align:left;
  display:inline;
}

.right {
   width: 39%;
   float:right;
   text-align:left;
   display:inline;
}

.container {
   width: 80%;
   margin:auto;
}

#first {

color: red;

}
```

```html
<html>
    <head>
          <Title>Content Alignment and CSS</Title>
          <link href='style.css' type='text/css' rel='stylesheet' />
    </head>
    <body>
          <div class='container'>
          <div class='left'>
                <h3>Few Indian Beverages</h3>
                <ul>
                      <li id='first'>Lassi</li>
                      <li>Chhaas</li>
                      <li>Aamras</li>
                      <li>Sugarcase juice</li>
                </ul>

          </div>
          <div class='right'>
                <h3>Popular Indian Beverages</h3>
                <ul>
                      <li>Coffee</li>
                      <li>Tea</li>
                      <li>Jal-jeera</li>
                      <li>Sharbat</li>
                </ul>
          </div>
          </div>
    </body>
</html>
```

Output:



Element selector

The element selector selects html elements based on the element name. When styling is applied based on the element name all elements that have the same name will be applied with the same styling.

```
p
{
text-align: center;
color: red;
}
```

For example the ruleset above will format all the <p> elements in the html document with the styling mentioned in the ruleset.

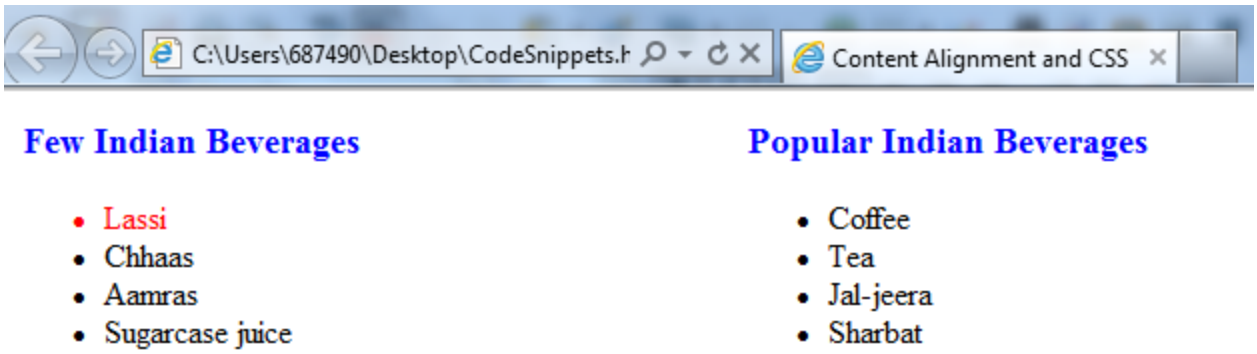
Objective: Use of tags to apply style in stylesheets

The styling is applied by using the html element name. In this case the style is applied using the style is applied to the html element h3.

```
.left {
  width: 39%;
  float:left;
  text-align:left;
  display:inline;
}

.right {
   width: 39%;
   float:right;
   text-align:left;
   display:inline;
}

.container {
   width: 80%;
   margin:auto;
}

#first {

color: red;

}

h3 {

color: blue;

}
```

```html
<html>
    <head>
          <Title>Content Alignment and CSS</Title>
          <link href='style.css' type='text/css' rel='stylesheet' />
    </head>
    <body>
          <div class='container'>
          <div class='left'>
                <h3>Few Indian Beverages</h3>
                <ul>
                      <li id='first'>Lassi</li>
                      <li>Chhaas</li>
                      <li>Aamras</li>
                      <li>Sugarcase juice</li>
                </ul>

          </div>
          <div class='right'>
                <h3>Popular Indian Beverages</h3>
                <ul>
                      <li>Coffee</li>
                      <li>Tea</li>
                      <li>Jal-jeera</li>
                      <li>Sharbat</li>
                </ul>
          </div>
          </div>
    </body>
</html>
```

Output



9.3.5 Inserting CSS into Html

There are 3 ways of inserting style sheets:

- External style sheet
- Internal style sheet
- Inline style

9.3.5.1 External style sheet

External style sheet is ideal when applying style to many pages. The styling code is maintained in a single file externally. The look of an entire website can be changed with changes done to just this single file.

An html page can link to an external style sheet using the <link> tag. The <link> tag is defined under the <head> tag.

```
<head>

<link rel="stylesheet" type="text/css" href="mystyle.css">

</head>
```

The import rule can also be used to import an external css file.

```
<head>

<style type="text/css">

    @import url(mystyle.css);
```

</style>

</head>

The external style sheet can be written in any text editor. The file should contain only the styling code and should not contain any html tags. The style sheet must be saved with a .css extension.

Objective: *To apply style using an external CSS file and then include it in the HTML page.*

Create a CSS file named style.css (or any other name of your choice) and write the code shown in the image below.

'.' in front of each tag defines a class and corresponding class is mentioned in the element used in HTML page.

style.css

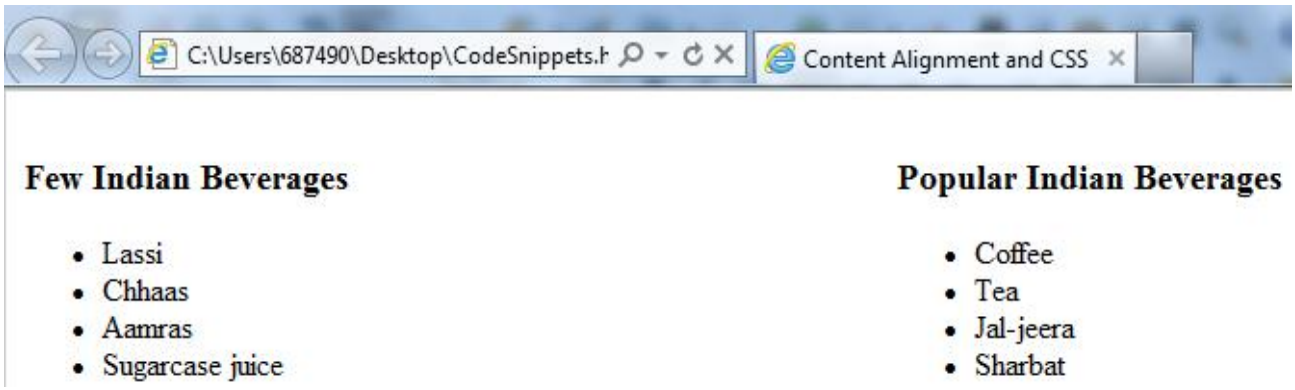
```
.left {
  width: 39%;
  float:left;
  text-align:left;
  display:inline;
}

.right {
   width: 39%;
   float:right;
   text-align:left;
   display:inline;
}

.container {
   width: 80%;
   margin:auto;
}
```

Create a HTML file in the same folder where the CSS file is created. HTML file should contain the following code.

The style sheet is imported in the head section using the html link tag.

```
<html>
    <head>
          <Title>Content Alignment and CSS</Title>
          <link href='style.css' type='text/css' rel='stylesheet' />
    </head>
    <body>
          <div class='container'>
          <div class='left'>
                <h3>Few Indian Beverages</h3>
                <ul>
                      <li>Lassi</li>
                      <li>Chhaas</li>
                      <li>Aamras</li>
                      <li>Sugarcase juice</li>
                </ul>

          </div>
          <div class='right'>
                <h3>Popular Indian Beverages</h3>
                <ul>
                      <li>Coffee</li>
                      <li>Tea</li>
                      <li>Jal-jeera</li>
                      <li>Sharbat</li>
                </ul>
          </div>
          </div>
    </body>
</html>
```

Output:



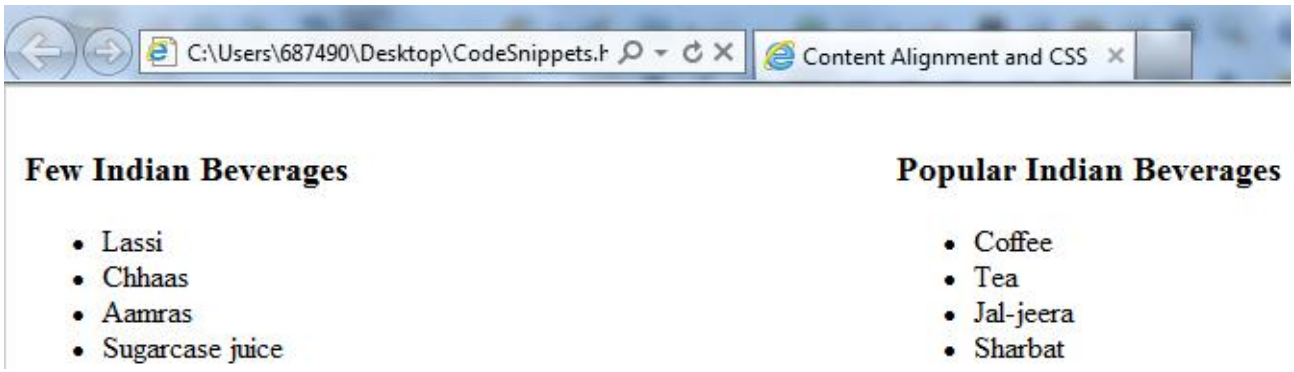
9.3.5.2 Internal style sheet

Internal style sheets are used when applying unique styles to single documents. Internal styles are written inside <head> tags using <style> tag.

```
<head>
<style>
.left
{
color:blue;
text-align:left;
font-family:"calibri";
}
</style>
</head>
```

Objective: *To apply internal style to div tags, so that two div tags are displayed side by side. The style applied is present inside style tags in the head section of HTML file.*

```
<html>
    <head>
        <Title>Content Alignment and CSS</Title>

        <style type='text/css'>
            .left {
              width: 39%;
              float:left;
              text-align:left;
              display:inline;
            }

            .right {
               width: 39%;
               float:right;
               text-align:left;
               display:inline;
            }

            .container {
               width: 80%;
               margin:auto;
            }

        </style>
    </head>
    <body>
        <div class='container'></div>
        <div class='left'>
            <h3>Few Indian Beverages</h3>
            <ul>
                <li>Lassi</li>
                <li>Chhaas</li>
                <li>Aamras</li>
                <li>Sugarcase juice</li>
            </ul>

        </div>
        <div class='right'>
            <h3>Popular Indian Beverages</h3>
            <ul>
                <li>Coffee</li>
                <li>Tea</li>
                <li>Jal-jeera</li>
                <li>Sharbat</li>
            </ul>
        </div>
        </div>
    </body>
</html>
```



Output:

9.3.5.3 Inline styles

Inline styles are written as part of the html tag itself. The style attribute of the html tag is used and can contain any css property. This way of styling should be used only sparingly and when the other two ways of inserting syles is not feasible since it is least flexible.

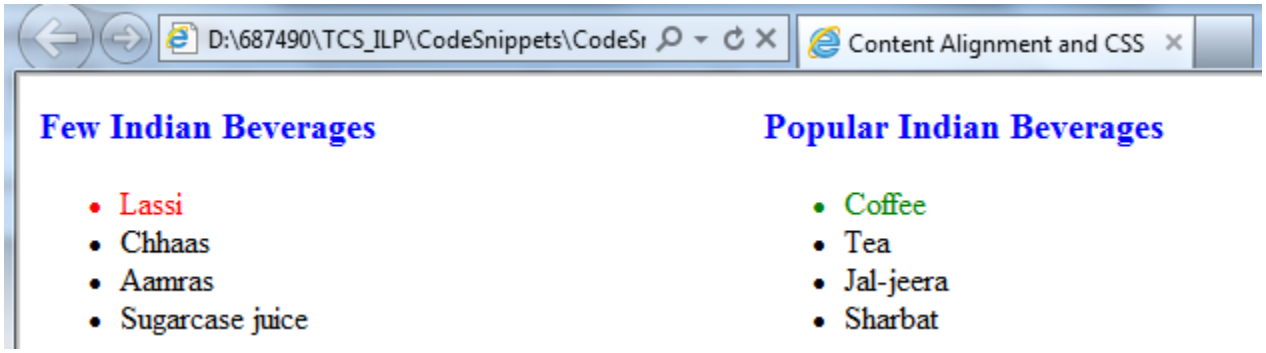
* This is a paragraph *

Objective: To apply styles at the html element/tag level or inline styling.

Html:

```html
<html>
    <head>
          <Title>Content Alignment and CSS</Title>
          <link href='style.css' type='text/css' rel='stylesheet' />
    </head>
    <body>
          <div class='container'>
          <div class='left'>
                <h3>Few Indian Beverages</h3>
                <ul>
                      <li id='first'>Lassi</li>
                      <li>Chhaas</li>
                      <li>Aamras</li>
                      <li>Sugarcase juice</li>
                </ul>

          </div>
          <div class='right'>
                <h3>Popular Indian Beverages</h3>
                <ul>
                      <li style='color: green'>Coffee</li>
                      <li>Tea</li>
                      <li>Jal-jeera</li>
                      <li>Sharbat</li>
                </ul>
          </div>
          </div>
    </body>
</html>
```

Output:



9.3.6 Overriding the Styles

Multiple styles, if applied to same element will reflect in following order:

1. Style attribute applied at element/tag level will have top most preference.
2. Style applied in the head section using style tag of HTML page will be preferred next.
3. External style will have the last preference.

What if I forcefully want to give preference to a style defined for an element in the external style sheet? Code shown below helps us to achieve it.

Objective: *Overriding the styles applied*

```
<html>
    <head>
          <Title>Content Alignment and CSS</Title>
          <link href='style.css' type='text/css' rel='stylesheet' />
    </head>
    <body>
          <div class='container'>
          <div class='left'>
                <h2>Few Indian Beverages</h2>
                <ul>
                      <li id='first'>Lassi</li>
                      <li>Chhaas</li>
                      <li>Aamras</li>
                      <li>Sugarcase juice</li>
                </ul>

          </div>
          <div class='right'>
                <h3>Popular Indian Beverages</h3>
                <ul>
                      <li style='color: green'>Coffee</li>
                      <li>Tea</li>
                      <li>Jal-jeera</li>
                      <li>Sharbat</li>
                </ul>
          </div>
          </div>
          <p>Above examples are taken for demonstrating the usage of style sheets and
          its implementation.</p>
    </body>
</html>
```

```
.left {
  width: 39%;
  float:left;
  text-align:left;
  display:inline;
}

.right {
   width: 39%;
   float:right;
   text-align:left;
   display:inline;
}

.container {
   width: 80%;
   margin:auto;
}

#first {

color: red;

}

h3 {

color: blue !important;

}

h2, h3, p {

color: grey;
font-weight: italics;
font-family: verdana;

}
```

Output

9.4 Reference Links

To know more on HTML and CSS, visit the URLs:

http://en.wikipedia.org/wiki/HTML

http://www.howtocreate.co.uk/tutorials/html/basics

http://www.htmlgoodies.com/tutorials/html_401/

http://www.w3schools.com/html/default.asp

http://html.net/tutorials/html/

http://www.tutorialspoint.com/html/what_is_new_in_html4.htm

http://www.w3resource.com/html/doctype-HTML.php

http://www.alternetwebdesign.com/htmltutorial/lesson3.htm

http://www.w3schools.com/css/DEFAULT.asp

http://learnlayout.com/